Lab 4 — Pendulum: data in Google Sheets

Class 1A — 4 November 2026

Group work, but everyone does everything on their own computer.
File name: `Lab4_surname` (for example `Lab4_Rossi`). **Hand in on Google Classroom.**

1 — Set up and enter the data

New **Blank** spreadsheet → Rename: `Lab4_surname` → File → Settings → Locale → **United Kingdom** then Save

Type decimals with a **point**. Type exactly this (numbers in A–D: yours, from the lab sheet; the values below are an example):

	A	B	C	D	E	F	G
1	L (cm)	t1 (s)	t2 (s)	t3 (s)	mean t (s)	T (s)	T^2 (s^2)
2	20	18.6	18.4				
3	30	22.4	22.7				
...	...	...	...	...			
8	...	...	...	...			

← row 1: headings with units

← one row per length, rows 2–8; t3 only if you have it

type here (from your lab sheet) — leave empty: formulas (step 2)

18.6 ✓ 18-Jun ×

a date, a time or text? Locale not set: fix it first.

2 — Formulas

Click the cell, type the formula, **Enter**. To select a range: click the first cell, **hold Shift**, click the last cell. Select `E2:G2`, then drag the small square down to row 8.

	A	B	C	D	E	F	G
1	L (cm)	t1 (s)	t2 (s)	t3 (s)	mean t (s)	T (s)	T^2 (s^2)
2	20	18.6	18.4		`=AVERAGE(B2:D2)`	`=E2/20`	`=F2^2`
3	30	22.4	22.7		`=AVERAGE(B3:D3)`	`=E3/20`	`=F3^2`
...	...	...	...	...	...	...	...
8	...	...	...	...	...	...	...

click `E2`, **Shift**, click `G2`

drag the small square down to row 8

E2 mean of the times **F2** time for one swing: $t/20$ **G2** T^2 (or `=F2*F2`)

Check: for $L = 100$ cm, $T \approx 2$ s, and T grows with L. Show 3 decimals in F and G (select `F2:G8`: click `F2`, **Shift**, click `G8`).

3 — Graph of T^2 against L (points only)

	A	B	C	D	E	F	G
1	L (cm)	t1	t2	t3	mean t	T (s)	T^2 (s^2)
2							
...		...	...	...	...	...	
8							

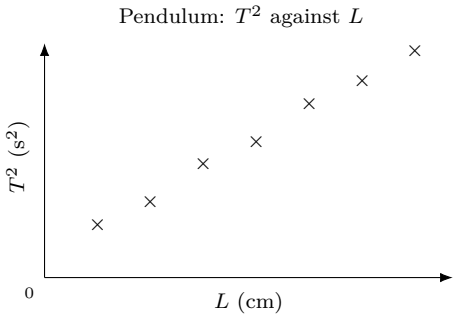
1. click `A1`, **Shift**, click `A8`

2. hold **Ctrl** (Cmd) and drag over `G1:G8`

Insert → **Chart**, then set the **Chart editor** like this:

Setup
Chart type:
Scatter chart
X-axis: **A**
Series: **G** only

Customize
Chart & axis titles:
title, `L (cm)`, `T^2 (s^2)`
Horizontal axis: Min = **0**
Vertical axis: Min = **0**

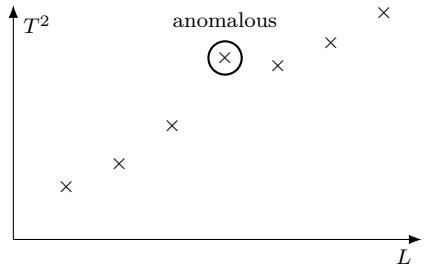


4 — Anomalous result (before drawing any line)

An **anomalous result** is a point that does not follow the pattern of the others. A line pulls towards it, so we look for it **first**. If T^2 is proportional to L, the ratio T^2/L is about the same for every length.

A	G	H
L (cm)	T^2 (s^2)	T^2/L (s^2/cm)
20	0.865	0.043
30	1.254	0.042
40	1.729	0.043
50	2.300	0.046
60	2.481	0.041
80	3.276	0.041
100	4.060	0.041

← clearly different

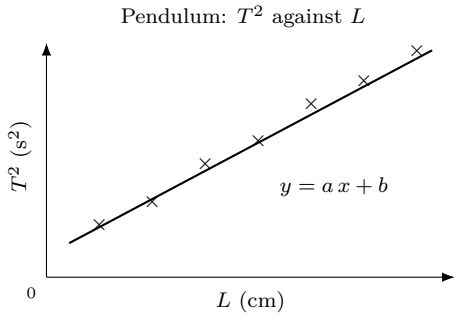


`H2: =G2/A2`, fill down to `H8`
(example numbers)

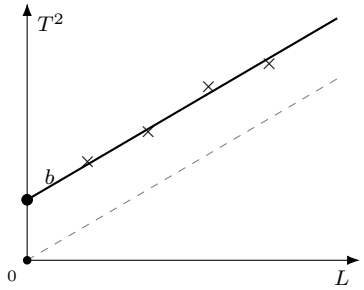
Point far from the others? Ratio different? → Check the table against your lab sheet → Still different? **Do not delete it** → Write: which L + one possible cause (swing miscounted, stopwatch late...)

5 — Trendline, and bonus: the intercept

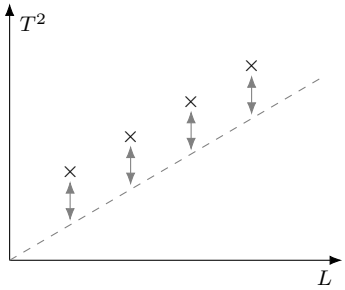
Customize → Series
☒ Trendline
Type: Linear
Label: **Use Equation**



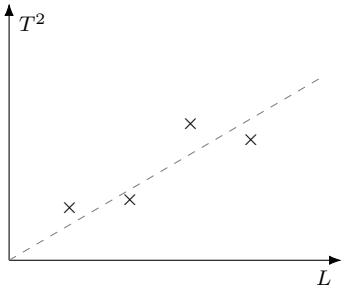
In $y = a\,x + b$, b is the **intercept**: the value of y where the line crosses the vertical axis, that is, y when $x = 0$. Here: the T^2 the line gives for $L = 0$. No length, no swing: we expect $b \approx 0$ (unit of b: s^2).



intercept b (dashed: expected, $b = 0$)



systematic: all points shifted by the same amount, $b \neq 0$



random: above and below the line

Write: your b, is it close to zero?, one possible systematic error (for example the ruler did not start exactly at the pivot).

6 — If there is time

How many swings keep the timing error under 1%? Estimate the uncertainty Δt of one timing from t_1 and t_2.

	F	I	J	K
row 1	T (s)	half difference (s)	uncertainty on T (%)	Δt (s)
row 2	···	`=ABS(B2-C2)/2`	`=$K$2/(20*F2)*100`	`=AVERAGE(I2:I8)`

`$` keeps `K2` fixed

Fill down `I2` to `I8` and `J2` to `J8`. $\dfrac{\Delta t}{20\,T} \times 100$ is the percentage uncertainty on T.

For which L is it below 1%? Which pendulum is hardest to time? Fill in "After the class" on your lab sheet.

7 — Hand in

Table, formulas and graph are on **your own** sheet → File name: `Lab4_surname` → Hand in on **Google Classroom**